Exp: 3 Study of different types of Network cables and crimping of cable with RJ45 connector

a) Understand different types cables

i) Unshielded Twisted Pair :

category 3 - 10 bPS Maximum Data transmission

category 5 - upto 100 MbPS

category 5e - 1 GbPS

Advantages :

* cheap in cost
* Easy to install as they have smaller overall diameter

Disadvantages :

* More Prone to Electromagnetic Interferenc

Use :

* 10Base-T Ethernet
* Fast Ethernet, Gigabit Ethernet
* Fast Ethernet, Gigabit Ethernet

ii) Shielded Twisted Pair (STP) cable :

category 6, 6a - 10 GbPS

SSTP - category 7 - 10GbPS

Advantage :

* Shielded
* Faster than UTP

Disadvantage :

* Expensive
* Greater Installation effort

iii) Coaxial cable :

RG-6
RG - 10 - 100 MbPS
59
RG 11

Advantage :

* High bandwith
* Immune to Interference
* Versatile

Disadvantage :

* Limited distance
* cost
* size in bulky

iv) Fibre optics cable :

Single mode Multi mode - 100 GbPS

Advantage :

* High speed
* High bandwidth

Disadvantage :

* Expensive
* Requires skilled installers

b) Make your own Ethernet cross-over cable / Straight cable :

Steps :

* To start construction of the device, begin by threading shields onto the cable
* Next, strip approxmately 1.5cm of cable shielding from both ends. The crimping tool has round area to complete this task
* After, you will need to untangle the wires ; there should be four "twisted pairs." Referencing back to sheet, arrange them from top to bottom. One ends should be arrangement A and other B

* Once the order is correct, bunch them together in line and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 Plug without messing up the order. To do so, hold the Plug with clip side facing away from you and have the gold pins facing towards you

* Next, push the cable right in. The notch at the end of the Plug needs to be just over the cable shielding and if it isn't, that means that you stripped off too much shielding. Simply snip cables back little more.

* After the wires are securely sitting inside the Plug, insert it into the crimping tool and push down

* Lastly, repeat for other end using diagram B to make crossover cables. Using diagram A to make a straight through cable

Result :

Thus, the study of different types of network cables and crimping of cable with RJ45 connector is shown